

Assignment ambulance location

In the Netherlands, ambulances should arrive within 15 minutes at the scene of an incident. To guarantee this, ambulances must be placed at strategic locations. The ambulances are usually stationed at base locations where they wait until an emergency call comes in. These base locations should be located such that the population that can be reached within 15 minutes is maximized. In addition, since every minute counts, we want to minimize the response time for the entire region.

The set of demand locations where people live is denoted by I and the demand for this location $i \in I$ is given by d_i , which represents the population at this demand location. The set of potential base locations is denoted by J . From this set, p locations can be chosen where one ambulance is stationed. The driving time between each base location $j \in J$ and demand location $i \in I$ is given by t_{ji} . When one of the p ambulances is stationed at a base location j for which $t_{ji} \leq 15$, then the population in demand location $i \in I$ can be reached within 15 minutes.

The assignment is to determine at which base locations the p ambulances should be stationed such that the total population that can be reached within 15 minutes is maximized and such that the average response time over all demand locations is minimized.

1. Come up with an intuitive method/algorithm to find a feasible and good solution to this problem. Implement and test this algorithm on the given data sets. This method will be problem specific and only relies on logical thinking.
2. Set up a general Integer Linear Programming model for this problem.
3. Solve this problem to optimality for the given data sets.
4. Expand the basic problem by adding more details, extending it or modifying it.